

Saket Upperla

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EDUCATION

University of Texas at Austin

Aug. 2023 – June 2027

B.S. Computer Science | **GPA:** 3.91/4.0

Austin, TX

- **Courses:** Data Structures, Discrete Math, Robot Learning, Multivariable Calculus, Computer Architecture, Linear Algebra, Operating Systems, Principles of Machine Learning

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, HTML, CSS

Frameworks: React.js, Next.js, Node.js, Express.js

Databases: MongoDB

Developer Tools: Git, Axios, Postman, JWT

EXPERIENCE

Undergraduate Researcher

Sept. 2024 – Present

RobIn Lab

Austin, TX

- Leading a team of student researchers in developing and optimizing Graph Inverse Reinforcement Learning (GraphIRL) code to enhance functionality and scalability.
- Designing depth perception algorithms to extend 2D graph abstractions into 3D, improving reward function robustness and enabling more accurate task progress modeling through temporal matching.
- Addressing critical software bugs and optimizing graph-based reinforcement learning pipelines to improve model reliability and accuracy, with a current focus on achieving measurable gains through repeated simulation of 3D objects performing identical tasks in diverse environments.
- Planning to open-source the GraphIRL framework to foster broader collaboration and accelerate research advancements in inverse reinforcement learning.

PROJECTS

Travel Diaries | MongoDB, Express.js, React.js, Node.js, JWT

December 2024

- Developed a full-stack travel story web application, enabling users to sign up, log in, and create personal travel stories with image uploads and travel dates
- Implemented search and filtering features, allowing users to find stories by keyword, filter by date range, and pin favorite stories for easy access
- Designed a secure backend using Node.js and Express, featuring JWT-based authentication, MongoDB for storing user profiles and travel stories, and RESTful APIs for CRUD operations and image uploads
- Built a dynamic and responsive front-end using React, integrating interactive UI components to enhance user experience and ensure seamless data flow between client and server

Storytunes | Next.js, Express.js, Spotify API, Axios

July 2024

- Developed a fully functional web application with a dynamic Next.js front-end, integrating the Spotify API to recommend theme-based songs for Instagram stories.
- Utilized Axios for efficient and streamlined communication with the Spotify API.
- Leveraged Express.js to host a web server, manage HTTP requests and routing, and facilitate interactions with the Spotify API.

ROS Object Detection | Python, ROS, YOLOv5

May 2024

- Developed and implemented a ROS (Robot Operating System) wrapper for the YOLOv5 object detection model, integrating it with an Intel RealSense D435 camera to enable real-time object detection in robotic applications
- Integrated a ROS publisher node that broadcasts detailed object information to other ROS nodes
- Created a custom dataset using Roboflow and applied transfer learning techniques to refine the object detection model for dining-related objects, improving accuracy of dining object detection by 4.2%.